continous_probability

Siddique ur Rehman

2026-11-05

```
#density at 5
dunif(5,min=0,max=10)
```

```
## [1] 0.1
```

```
punif(5,min=0,max=10)
```

```
## [1] 0.5
```

```
qunif(0.95,min=0,max = 10)
```

```
## [1] 9.5
```

```
random_numbers=runif(10,min = 0,max = 10)
random_numbers
```

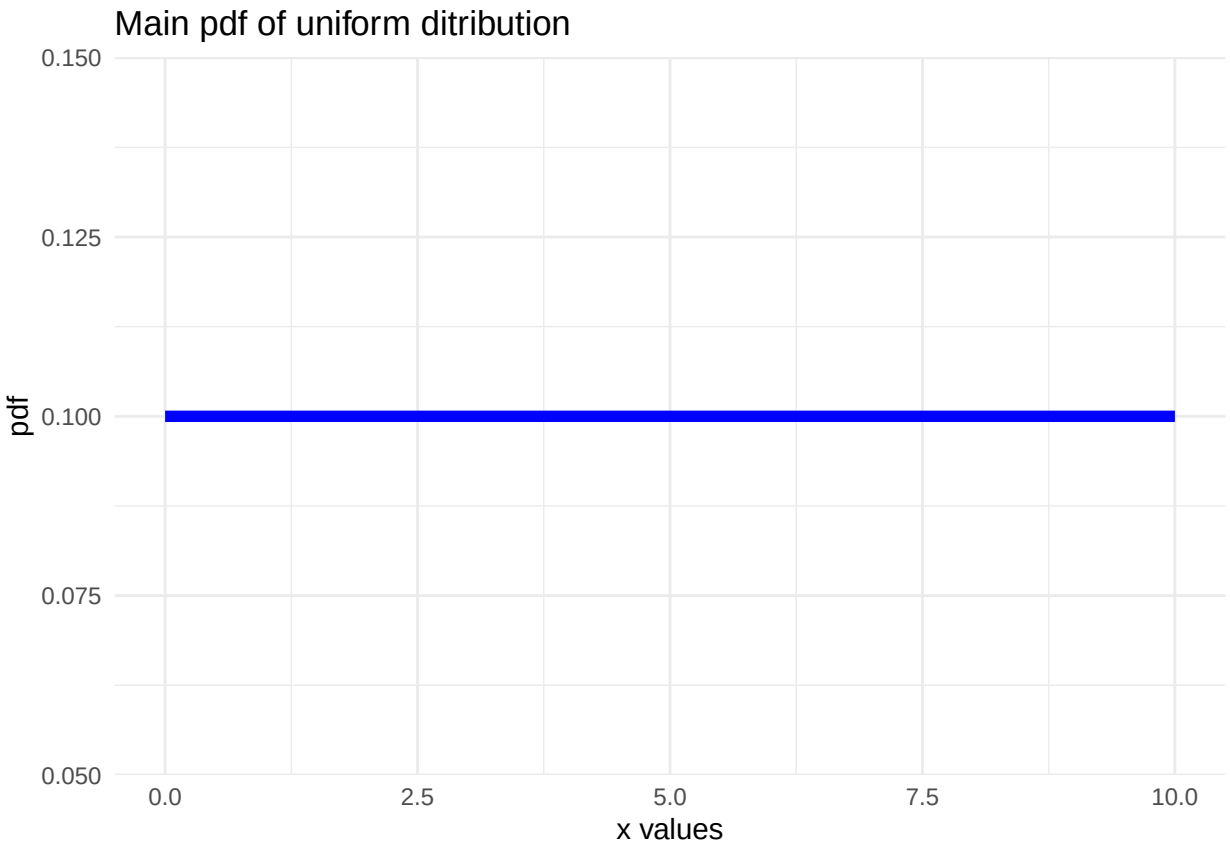
```
##  [1] 0.4665479 1.2581329 2.0545927 6.6977093 0.7772974 1.7697048 1.9925864
##  [8] 6.2329786 2.8760289 5.6043293
```

```
library(ggplot2)

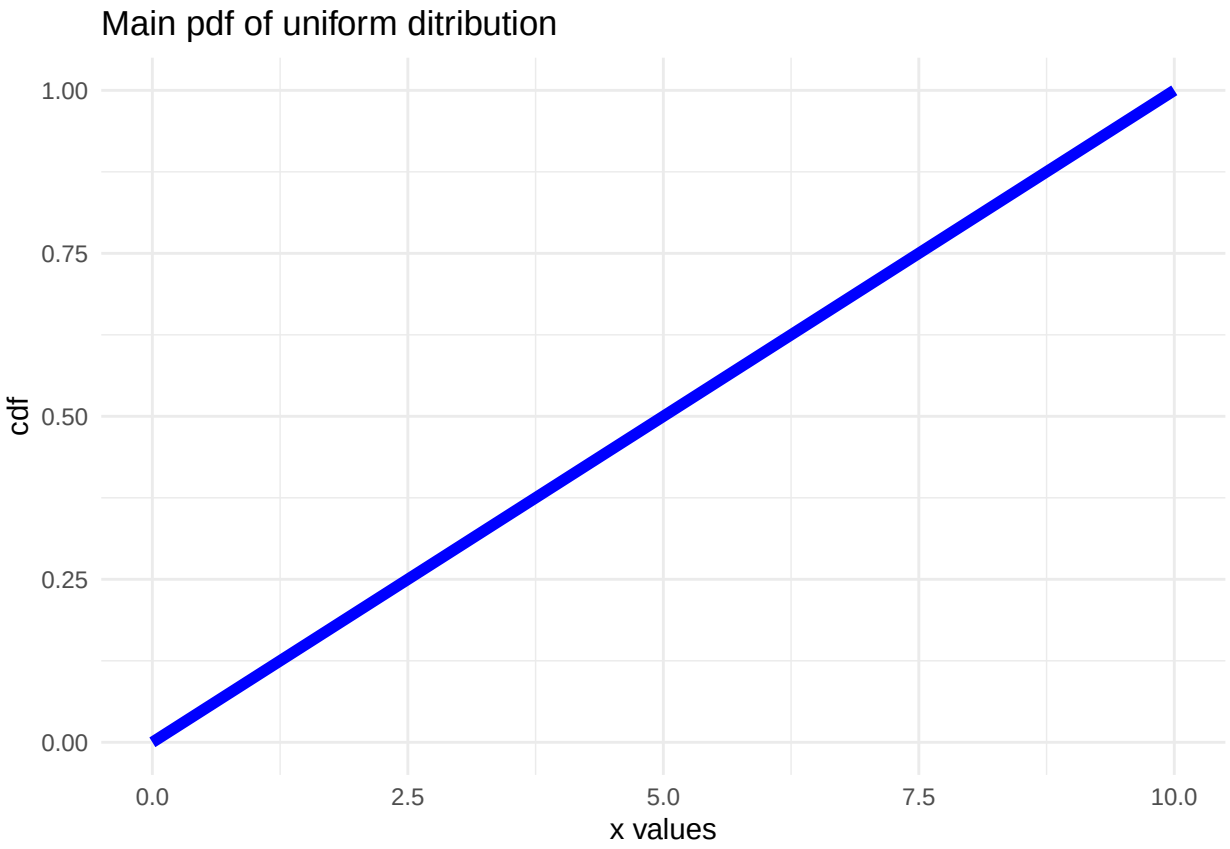
x<-seq(0,10,length.out=1000)
pdf<-dunif(x,min=0,max = 10)

unfirom_data<-data.frame(
  x=x,
  pdf=pdf
)

ggplot(unfirom_data,aes(x,pdf))+
  geom_line(colour = 'blue',linewidth = 2)+
  labs(
    x="x values",
    y="pdf",
    title = "Main pdf of uniform ditribution"
  )+
  theme_minimal()
```




```
cdf<-punif(x,min = 0,max = 10)
uniform_cdf<-data.frame(
  x=x,
  cdf=cdf
)
ggplot(uniform_cdf,aes(x,cdf))+
  geom_line(colour = 'blue',linewidth = 2)+
  labs(
    x="x values",
    y="cdf",
    title = "Main pdf of uniform ditribution"
  )+
  theme_minimal()
```




```
dnorm(60,mean = 50,sd=10)
```

```
## [1] 0.02419707
```

```
pnorm(60,mean=50,sd=10)
```

```
## [1] 0.8413447
```

```
pnorm(60,mean=50,sd=10)
```

```
## [1] 0.8413447
```

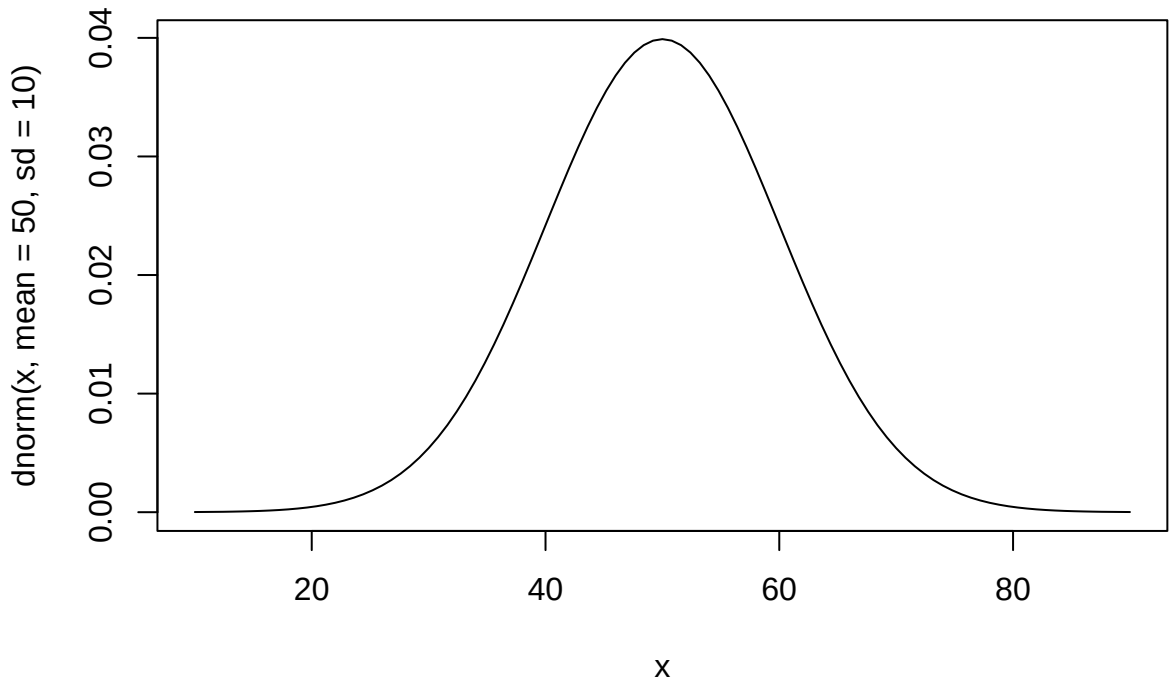
```
qnorm(0.95,mean = 50,sd=10)
```

```
## [1] 66.44854
```

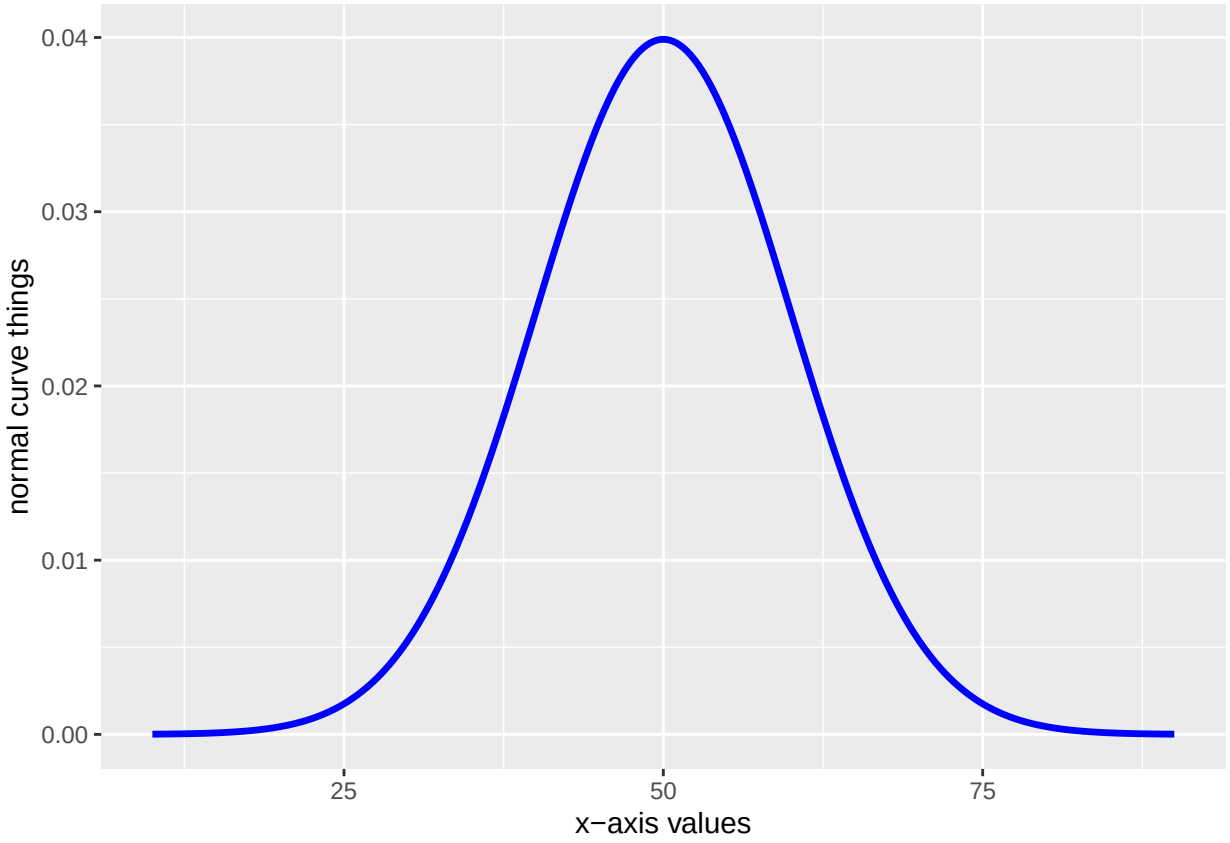
```
rnorm(10,mean=50,sd=10)
```

```
##  [1] 46.33650 44.93842 70.35952 47.52934 48.49685 58.98273 68.95613 43.36977
##  [9] 38.37244 50.31377
```

```r
x<-60
curve(dnorm(x,mean=50,sd=10),
      from = 10,to=90)
```




```r
library(ggplot2)
x_values<-seq(10,90,length.out=1000)
pdf_norm<-dnorm(x_values,mean = 50,sd=10)
my_data<-data.frame(
  x=x_values,
  pdf_norm=pdf_norm
)
ggplot(my_data,aes(x,pdf_norm))+
  geom_line(color='blue',linewidth=1.2)+
  labs(
    x="x-axis values",
    y="normal curve things"
  )
```




```
dnorm(1.96)
```

```
## [1] 0.05844094
```

```
pnorm(1.96)
```

```
## [1] 0.9750021
```

```
pnorm(1.96,lower.tail = FALSE)
```

```
## [1] 0.0249979
```

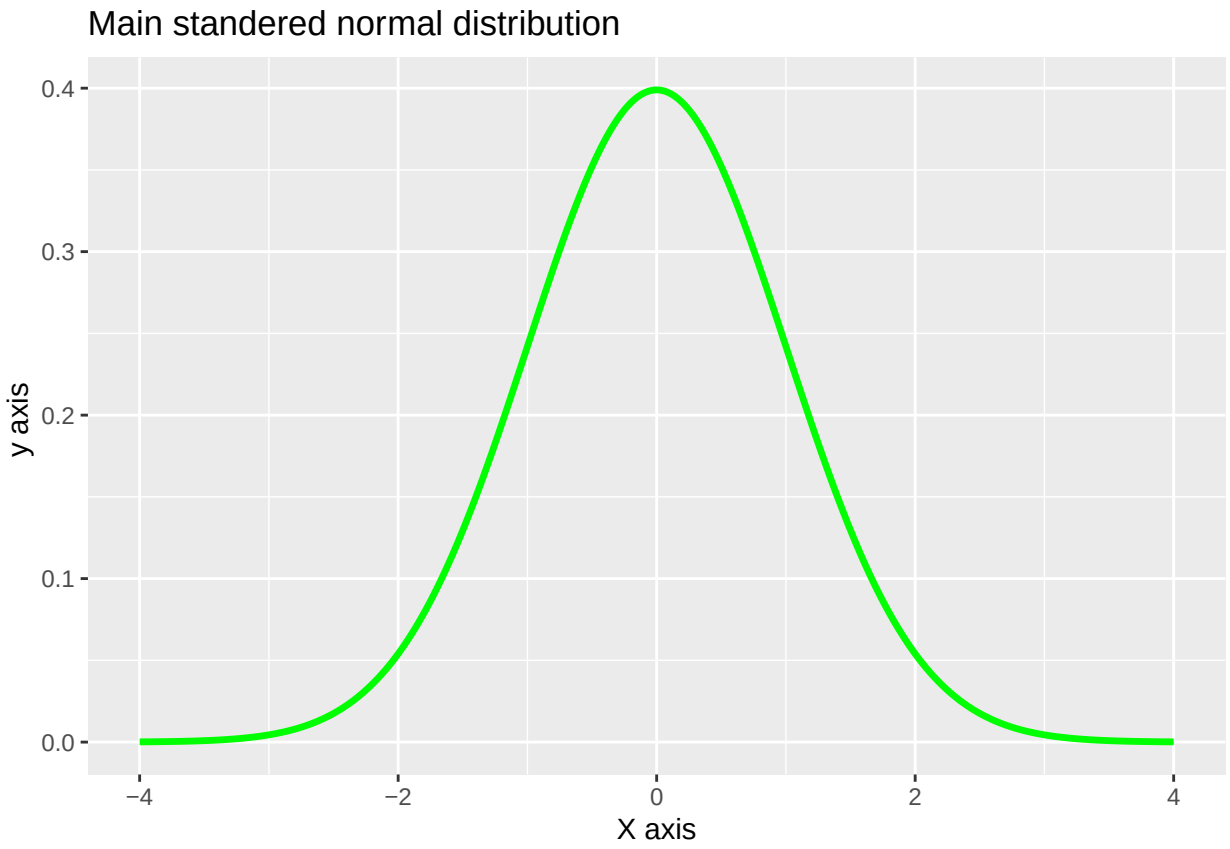
```
qnorm(0.976)
```

```
## [1] 1.977368
```

```
rnorm(10)
```

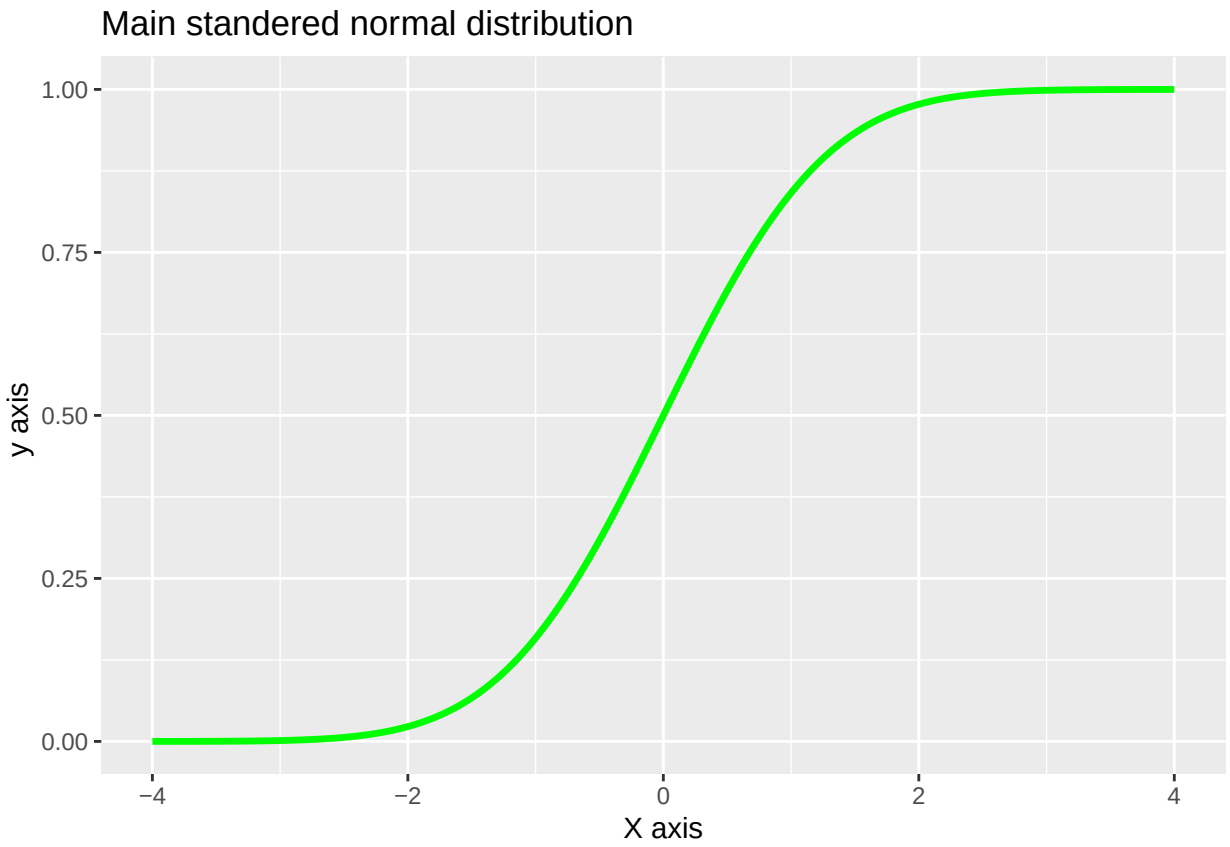
```
##  [1] -1.37855628 -0.09160599 -0.10368165  0.37501205 -1.47509164  0.62527288
##  [7] -1.38149585 -0.13237836 -0.14014861  0.01522129
```

```r
std_norm<-seq(-4,4,length.out=1000)
pdf_std_norm<-dnorm(std_norm)
std_data<-data.frame(
  x=std_norm,
  pdf=pdf_std_norm
)
ggplot(std_data,aes(x,pdf))+
  geom_line(color='green',linewidth = 1.2)+
  labs(
    x="X axis",
    y="y axis ",
    title = "Main standered normal distribution"
  )
```




```r
std_norm<-seq(-4,4,length.out=1000)
cdf_std_norm<-pnorm(std_norm)
std_data<-data.frame(
  x=std_norm,
  cdf=cdf_std_norm
)
ggplot(std_data,aes(x,cdf))+
  geom_line(color='green',linewidth = 1.2)+
  labs(
    x="X axis",
    y="y axis ",
```

```
    title = "Main standered normal distribution"
  )
```



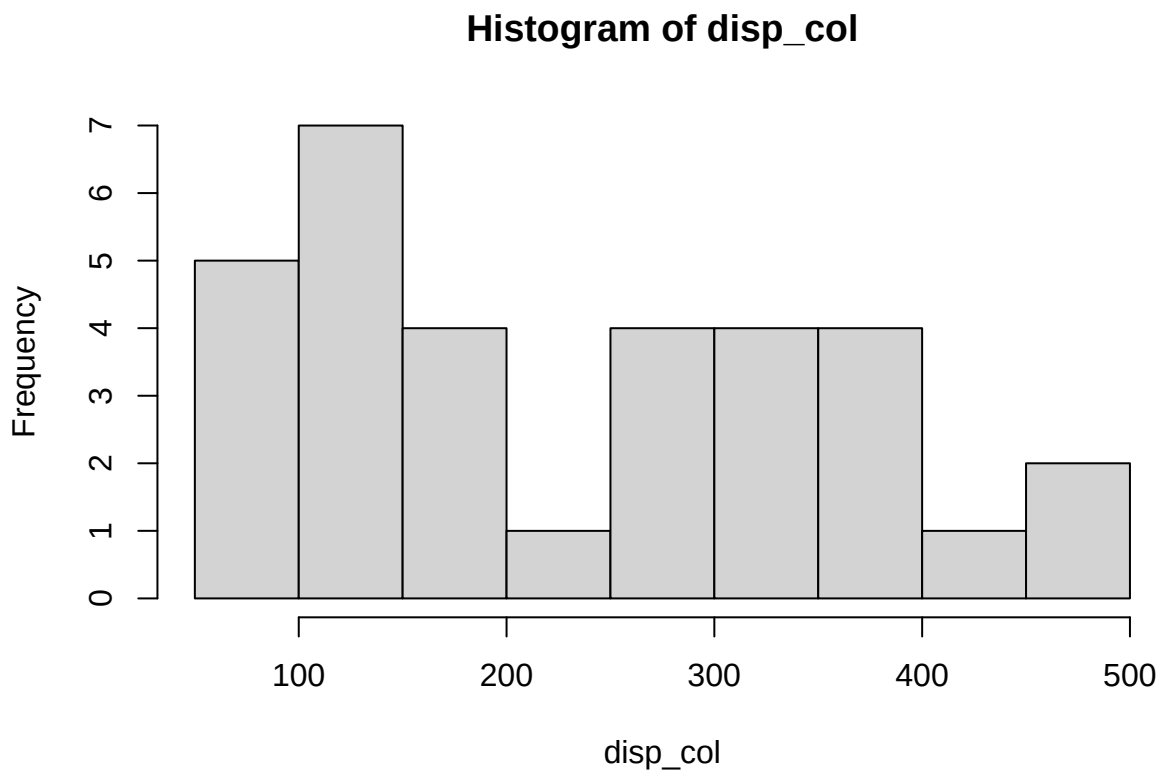

```
##Checking normality
```

```
datasets::mtcars
```

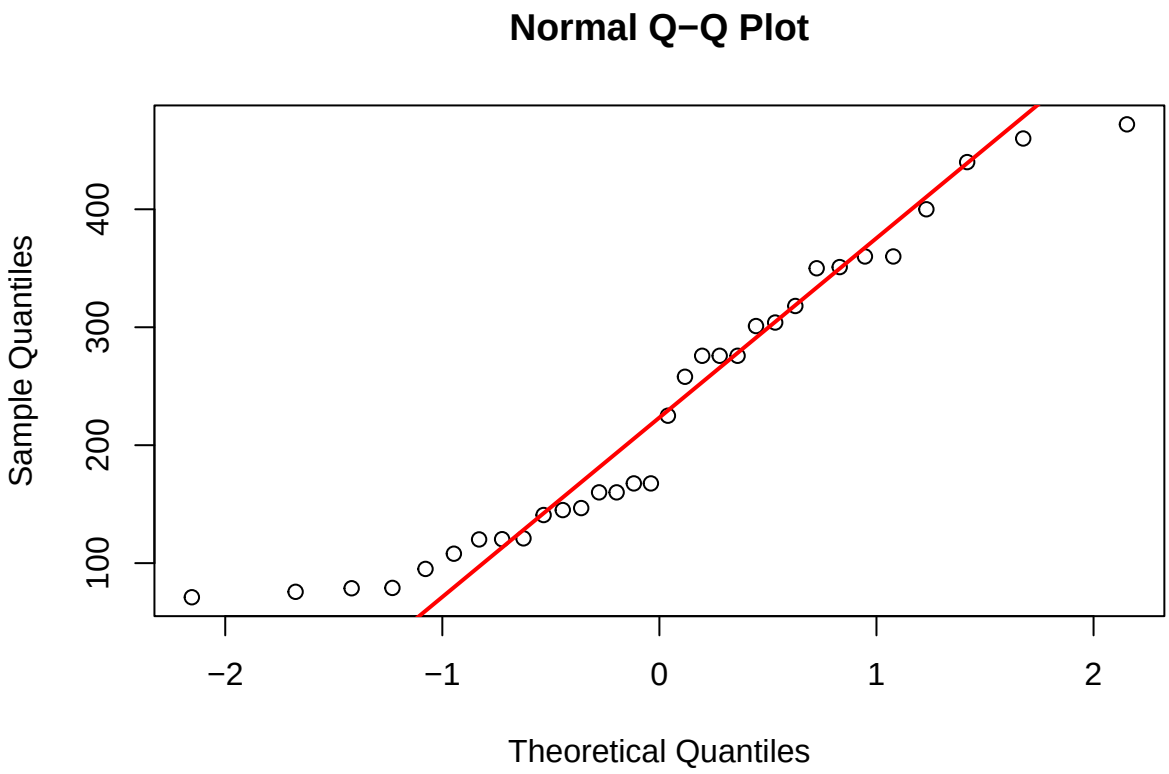
```
##                      mpg cyl  disp  hp drat    wt  qsec vs am gear carb
## Mazda RX4           21.0   6 160.0 110 3.90 2.620 16.46  0  1    4    4
## Mazda RX4 Wag       21.0   6 160.0 110 3.90 2.875 17.02  0  1    4    4
## Datsun 710          22.8   4 108.0  93 3.85 2.320 18.61  1  1    4    1
## Hornet 4 Drive      21.4   6 258.0 110 3.08 3.215 19.44  1  0    3    1
## Hornet Sportabout   18.7   8 360.0 175 3.15 3.440 17.02  0  0    3    2
## Valiant             18.1   6 225.0 105 2.76 3.460 20.22  1  0    3    1
## Duster 360          14.3   8 360.0 245 3.21 3.570 15.84  0  0    3    4
## Merc 240D           24.4   4 146.7  62 3.69 3.190 20.00  1  0    4    2
## Merc 230            22.8   4 140.8  95 3.92 3.150 22.90  1  0    4    2
## Merc 280            19.2   6 167.6 123 3.92 3.440 18.30  1  0    4    4
## Merc 280C           17.8   6 167.6 123 3.92 3.440 18.90  1  0    4    4
## Merc 450SE          16.4   8 275.8 180 3.07 4.070 17.40  0  0    3    3
## Merc 450SL          17.3   8 275.8 180 3.07 3.730 17.60  0  0    3    3
## Merc 450SLC         15.2   8 275.8 180 3.07 3.780 18.00  0  0    3    3
## Cadillac Fleetwood  10.4   8 472.0 205 2.93 5.250 17.98  0  0    3    4
## Lincoln Continental 10.4   8 460.0 215 3.00 5.424 17.82  0  0    3    4
## Chrysler Imperial   14.7   8 440.0 230 3.23 5.345 17.42  0  0    3    4
```

```
## Fiat 128           32.4   4  78.7  66 4.08 2.200 19.47  1  1    4    1
## Honda Civic        30.4   4  75.7  52 4.93 1.615 18.52  1  1    4    2
## Toyota Corolla     33.9   4  71.1  65 4.22 1.835 19.90  1  1    4    1
## Toyota Corona      21.5   4 120.1  97 3.70 2.465 20.01  1  0    3    1
## Dodge Challenger   15.5   8 318.0 150 2.76 3.520 16.87  0  0    3    2
## AMC Javelin        15.2   8 304.0 150 3.15 3.435 17.30  0  0    3    2
## Camaro Z28         13.3   8 350.0 245 3.73 3.840 15.41  0  0    3    4
## Pontiac Firebird   19.2   8 400.0 175 3.08 3.845 17.05  0  0    3    2
## Fiat X1-9          27.3   4  79.0  66 4.08 1.935 18.90  1  1    4    1
## Porsche 914-2      26.0   4 120.3  91 4.43 2.140 16.70  0  1    5    2
## Lotus Europa       30.4   4  95.1 113 3.77 1.513 16.90  1  1    5    2
## Ford Pantera L     15.8   8 351.0 264 4.22 3.170 14.50  0  1    5    4
## Ferrari Dino       19.7   6 145.0 175 3.62 2.770 15.50  0  1    5    6
## Maserati Bora      15.0   8 301.0 335 3.54 3.570 14.60  0  1    5    8
## Volvo 142E         21.4   4 121.0 109 4.11 2.780 18.60  1  1    4    2
```

```r
disp_col<-mtcars$disp
hist(disp_col)
```




```r
qqnorm(disp_col,main = "Normal Q-Q Plot")
qqline(disp_col, col = "red", lwd = 2)
```




```
shapiro.test(disp_col)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  disp_col
## W = 0.92001, p-value = 0.02081
```

```
library(nortest)

ad.test(disp_col)
```

```
##
##  Anderson-Darling normality test
##
## data:  disp_col
## A = 0.8745, p-value = 0.02211
```

```
population<-1:100

simple_random<-sample(population,10,replace = FALSE)
print(simple_random)
```

```
##  [1] 19  9 31  4 57 65 20 73 70 91
```